Green-Edge-Cloud: Routing without Routers via Confidence Calibration

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Abstract

Large language models (LLMs) deliver strong reasoning capabilities but incur prohibitive energy costs when deployed in the cloud. While edge deployment offers a low-power alternative, small models struggle with complex reasoning tasks and are prone to overconfident hallucinations. We propose a Green-Edge-Cloud collaborative framework that dynamically routes queries between a 1.5B edge model and a 72B cloud model. We introduce a Routing-Oriented Chain-of-Thought (CoT) distillation method that calibrates the edge model's intrinsic sequence likelihood to align with correctness, improving epistemic reliability. This enables reliable routing decisions via confidence ranking, without auxiliary routing policies or additional training. We further perform hardware-level measurements using `CodeCarbon`, revealing non-trivial idle energy during offloading, a factor often overlooked in prior work. By explicitly modeling this effect, our framework achieves a superior energy–accuracy trade-off. By offloading only 15.2% of queries, the system achieves 79.8% accuracy on GSM8K while reducing energy consumption by 83% compared to a cloud-only setup.

1. Introduction

Large language models (LLMs) deliver strong reasoning but incur prohibitive energy and latency in the cloud (Samsi et al., 2023; Niu et al., 2025). While edge deployment offers a low-power alternative, small models often fail on complex tasks, creating a tension between efficiency and performance.

A natural solution is to route queries between edge and cloud models. However, existing approaches treat routing as a separate learning problem, relying on auxiliary routing policies (e.g., DRL or MAB) (Yuan et al., 2024; Xu et al., 2026) or fine-grained model partitioning (Fan et al., 2025), both of which introduce additional system complexity and overhead.

In this paper, we challenge the necessity of explicit routing policies. We show that effective query routing does not require any auxiliary policy, but can emerge directly from a model-intrinsic confidence signal when properly calibrated.

We therefore reformulate routing as an *uncertainty calibration problem*. While small LLMs are typically overconfident, we show that aligning confidence with correctness enables reliable routing decisions.

Based on this insight, we propose the **Green-Edge-Cloud** framework, where a small edge model acts as an intrinsic router without any auxiliary components. We introduce a *Routing-Oriented Chain-of-Thought (CoT) Distillation* method that aligns reasoning quality with confidence estimation, enabling a simple, lightweight threshold-based decision rule for routing.

We further highlight the importance of system-level evaluation by explicitly modeling edge idle energy during cloud offloading, a factor often overlooked in prior work.

Our main contributions are summarized as follows:

- **Routing without routers.** We show that routing can be reduced to model-intrinsic uncertainty estimation, eliminating the need for auxiliary routing policies.
- **Calibration via reasoning distillation.** We propose a CoT-based distillation method that aligns confidence with correctness, enabling reliable zero-shot routing.
- **System-aware energy modeling.** We perform hardware-level energy profiling with `CodeCarbon`, explicitly accounting for edge idle energy.
- **Improved accuracy–energy trade-off.** Our method achieves a superior Pareto frontier on GSM8K, reaching 79.8% accuracy while reducing energy by 83% with only 15.2% offloading.

2. Related Work

LLM Routing and Collaborative Execution. Traditional routing approaches focus on cloud-centric API selection

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(e.g., FrugalGPT (Chen et al., 2024), RouteLLM (Ong et al., 2024)), which introduce additional inference overhead and require specialized preference data for router training. Recent advancements in top-tier venues frame routing as a complex decision-making process, utilizing in-context learned representations (Wang et al., 2026) or formulating it as a contextual bandit problem under budget constraints (Panda et al., 2025). While effective, these approaches often introduce additional system complexity or require auxiliary routing modules. In contrast, our Green-Edge-Cloud framework relies on a zero-shot, model-intrinsic confidence signal, completely avoiding both trained external routers and intermediate state transfer.

Confidence Calibration for Cost-Efficient Reasoning. Leveraging model uncertainty for adaptive computation is an emerging frontier in LLM collaboration. Recent work such as COREA (Zhang et al., 2026) utilizes reinforcement learning to fine-tune small language models for confidence estimation, triggering large model assistance when the verbalized confidence is low. However, verbalized confidence can still be susceptible to hallucination. Our approach fundamentally differs by reformulating routing as an intrinsic probability calibration problem. By employing Routing-Oriented Chain-of-Thought (CoT) distillation, we naturally align the length-normalized sequence likelihood with reasoning correctness, yielding a highly reliable decision boundary without requiring complex RLHF pipelines.

Green AI and Edge-Cloud Inference. Inference overwhelmingly dominates the energy footprint of LLM deployment (Niu et al., 2025; Samsi et al., 2023). To mitigate computational bottlenecks, recent edge-cloud collaborative frameworks explore bandwidth-efficient speculative decoding (Bhattacharjee et al., 2025) to reduce latency. However, prior system-level evaluations often rely on theoretical proxy metrics (e.g., FLOPs) or isolated active computation time. In this work, we highlight the necessity of end-to-end accounting. By directly measuring hardware energy via `CodeCarbon`, we explicitly model the active tail and idle energy of edge devices during cloud offloading, providing a rigorous and physically grounded assessment of Green AI systems.

3. Methodology

Unlike heuristic routing based on shallow features (e.g., prompt length), we base routing decisions on model uncertainty. We show that calibrated token-level likelihood serves as a reliable proxy for reasoning difficulty. The Green-Edge-Cloud framework consists of two coupled components: a routing-oriented distillation process that calibrates the edge model, and a zero-shot routing mechanism that dynamically allocates computation.

3.1. Routing-Oriented CoT Distillation

To enable the small edge model (Student) to act as a reliable gatekeeper, it must be properly calibrated—meaning its confidence should correlate well with its correctness. We utilize Chain-of-Thought (CoT) distillation to transfer reasoning capabilities from the 72B Teacher model to the 1.5B Student model. Given queries X, we generate rationales R and answers Y using the Teacher. The Student is then fine-tuned by minimizing the standard negative log-likelihood (NLL) loss over the generated reasoning path:

$$\mathcal{L}_{CoT} = -\sum_{i=1}^{|R \oplus Y|} \log P_\theta(t_i | x, t_{<i}) \tag{1}$$

where θ denotes the parameters of the edge model, $x \in X$ is the input query, and t_i represents the tokens in the concatenated sequence of rationale and answer.

This process not only improves reasoning accuracy but also reshapes the student's confidence distribution. In standard direct-answer fine-tuning, models are prone to "shortcut learning," often assigning artificially high probabilities to incorrect final tokens due to memorized patterns.

By forcing the model to articulate intermediate reasoning steps via CoT distillation, the probability mass is distributed across the entire logical chain. Consequently, when the model lacks the requisite knowledge for a complex task, the accumulated token-level uncertainty gracefully propagates through the sequence. This mitigates overconfidence and aligns sequence likelihood with correctness, yielding a reliable routing signal and improved epistemic reliability through uncertainty accumulation.

3.2. Confidence Estimation and Uncertainty Calibration

A reliable routing signal must reflect the edge model's internal uncertainty (acting as a proxy for epistemic uncertainty) rather than raw generation likelihood. However, the autoregressive joint probability of a sequence is inherently length-biased, making it unsuitable for comparing responses with varying reasoning depths.

To obtain a stable, length-invariant uncertainty proxy, we define the sequence-level confidence as the normalized geometric mean of token probabilities:

$$\mathcal{C}(\mathbf{y}) = \exp\left(\frac{1}{N}\sum_{i=1}^{N} \log P(y_i \mid y_{<i}, \mathbf{x})\right) \tag{2}$$

This formulation transforms the autoregressive likelihood into a per-token confidence measure, which empirically correlates with correctness after calibration (Section 4.3). Importantly, $\mathcal{C}(\mathbf{y})$ serves as a model-intrinsic uncertainty signal, enabling routing decisions without auxiliary predictors.

During inference, the edge model generates a response and

computes $\mathcal{C}(\mathbf{y})$ on-device. Queries with $\mathcal{C}(\mathbf{y}) \geq \tau$ are handled locally, while low-confidence cases are offloaded to the cloud model. This aligns routing with uncertainty-aware decision-making rather than heuristic filtering.

4. Experiments

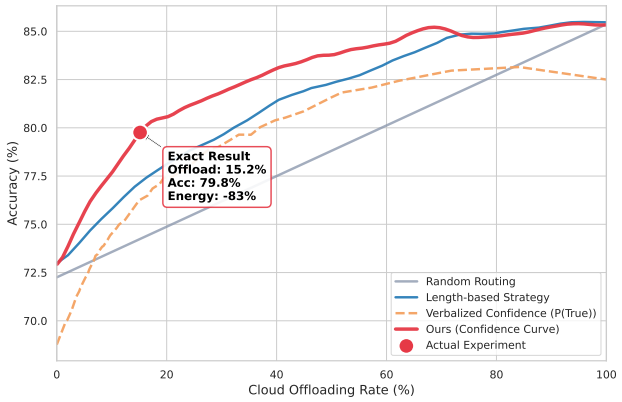


Figure 1. **Computation Cost vs. Accuracy Trade-off.** Our proposed Green-Edge-Cloud framework (Red curve) consistently dominates Random (Grey) and Length-based (Blue) routing baselines. The exact experimental result (red dot) achieves **79.8% accuracy** with an **83% energy reduction** compared to the cloud-only baseline.

Table 1. Overall performance on GSM8K and SVAMP. Energy consumption is normalized to the respective edge-only baseline.

Dataset	Strategy	Params	Offload	Acc.	Norm. Energy
GSM8K	Cloud-Only	72B	100.0%	82.5%	60.0×
	Edge-Only	1.5B	0.0%	72.3%	1.0×
	Ours	1.5B/72B	**15.2%**	**79.8%**	**9.95**×
SVAMP	Cloud-Only	72B	100.0%	91.3%	57.9×
	Edge-Only	1.5B	0.0%	82.0%	1.0×
	Ours	1.5B/72B	**20.0%**	**86.7%**	**19.5**×

4.1. Experimental Setup

Models, Hardware, and Dataset. We evaluate on two mathematical reasoning benchmarks (GSM8K and SVAMP). To simulate a realistic Edge-Cloud paradigm, we deploy `Qwen2.5-1.5B-Instruct` on a single RTX 3090 GPU as the local edge node. This setup effectively emulates the massive computational and energy disparity between a constrained local server and a distributed cloud backend. The cloud teacher model, `Qwen2.5-72B-Instruct` (GPTQ-Int4), is deployed across two RTX 3090 GPUs utilizing tensor parallelism ($TP = 2$) to accommodate its memory footprint. All inference pipelines are built upon the `vLLM` framework with `float16` precision. We observe negligible variance ($< 0.5\%$) across repeated runs.

Baselines. We compare our confidence-based routing against two standard heuristics and a recent contemporary approach:

- *Random Routing:* Queries are uniformly offloaded to the cloud with a predefined probability p.
- *Length-based Routing:* Queries are offloaded if the input prompt length exceeds a token threshold L, assuming longer queries are inherently more complex.
- *Verbalized Confidence (P(True)):* Inspired by recent collaborative reasoning frameworks (Zhang et al., 2026), we prompt the edge model to explicitly self-evaluate its proposed answer (i.e., answering "Yes" or "No" to whether it is correct). The generated probability of the "Yes" token is utilized as the routing confidence.

Empirical Energy Measurement. Instead of relying on theoretical FLOPs, we trace hardware power draw using `CodeCarbon`. To formalize the routing efficiency, the average per-query energy E_{avg} is modeled as a function of the offloading rate r:

$$E_{avg} = (1-r)E_{edge} + r(E_{edge} + E_{cloud} + P_{idle}^{edge} \cdot T_{cloud}) \quad (3)$$

where E_{edge} and E_{cloud} are the active inference energies, and P_{idle}^{edge} accounts for the edge device's active tail energy (e.g., keeping network modules active) while waiting for the cloud response. While our cascade paradigm requires the edge model to complete generation before offloading, this closed-loop formulation reveals its economic viability: the energy cost of the 1.5B edge model is orders of magnitude smaller than the 72B cloud model ($E_{edge} \ll E_{cloud}$). Consequently, the upfront local computation is highly economical, mathematically explaining the 83% total energy reduction compared to unconditional cloud routing.

4.2. Main Results

Pareto Efficiency. The Pareto frontier in Figure 1 is obtained by retrospectively sweeping the confidence threshold τ across the full distribution of edge model log-probabilities (from 0 to 1). As illustrated, our confidence-based routing strategy (Red curve) consistently dominates all baselines across the full cost spectrum. Notably, while Verbalized Confidence (P(True)) has shown great potential in larger models, it underperforms even the simple length-based heuristic in this highly constrained edge setting. This suggests that while small-capacity models (e.g., 1.5B) can generate answers, they may still lack the meta-cognitive capacity required for reliable self-evaluation, occasionally leading to overconfident hallucinations. By extracting the intrinsic sequence likelihood $\mathcal{C}(y)$ conditioned on CoT distillation, our framework gracefully bypasses this limitation, capturing the true epistemic uncertainty before explicit verbalization.

System-Level Performance. To evaluate a concrete deployment scenario without evaluation leakage, the routing threshold τ is selected on a randomly sampled subset of

the training split and fixed across all test queries. The reported operating point ($r = 15.2\%$) is strategically selected as the knee point of the Pareto frontier, which provides the best marginal gain in accuracy per unit of energy cost. As detailed in Table 1, this configuration successfully recovers the majority of the accuracy drop, achieving 79.8%. In our experiments, the normalized energy consumption is reduced to 9.95× relative to the edge baseline, leading to an 83% absolute energy saving compared to the cloud-only deployment.

4.3. Reliability of the Confidence Signal

We analyze the relationship between the intrinsic confidence score $C(y)$ and the actual problem-solving accuracy. While we do not explicitly optimize standard absolute calibration metrics (e.g., Expected Calibration Error), we empirically observe that our distilled confidence signal exhibits a consistent alignment with correctness. This suggests that routing can be effectively driven by relative uncertainty rather than absolute probability estimates.

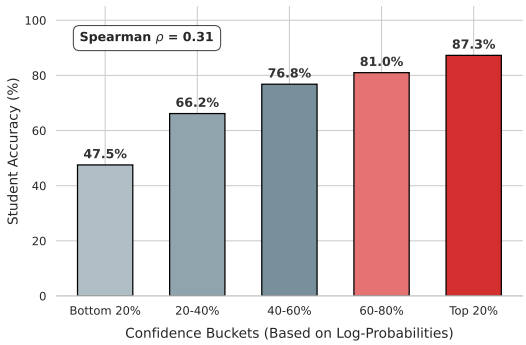


Figure 2. **Accuracy vs. Confidence Buckets.** Queries are sorted by the student model's intrinsic confidence and divided into quintiles. The clear monotonic trend indicates that the confidence signal provides a reliable *ranking* of query difficulty, which is sufficient for effective routing decisions.

As shown in Figure 2, higher confidence thresholds consistently correspond to higher success rates. Crucially, we argue that for dynamic edge-cloud routing, a well-calibrated *ranking* of uncertainty is actually more critical than absolute probability calibration. We emphasize that our objective in this work is not to achieve absolute probability calibration (e.g., minimizing Expected Calibration Error), but rather to ensure a consistent ranking of uncertainty that aligns with task correctness. This monotonic trend indicates that the confidence signal induces a consistent ordering over query difficulty, which is sufficient to define an effective decision boundary for routing. This empirical alignment effectively validates our design in Section 3.2, demonstrating that the length-normalized sequence likelihood provides a reliable, model-intrinsic proxy for epistemic uncertainty in routing decisions.

4.4. Ablation Study: The Impact of CoT Distillation

To isolate the effect of our Routing-Oriented CoT Distillation, we compare the distilled model with its vanilla counterpart and other small models.

Table 2. Ablation on Edge Models. "Vanilla" indicates the original off-the-shelf model without our confidence calibration distillation. The DeepSeek model is constrained to 1024 max tokens to simulate an edge-friendly budget.

Edge Student	Cloud Teacher	Offload	Accuracy	Energy Reduction
Gemma-2-2B (Vanilla)	Qwen2.5-72B	15.2%	64.9%	**84.1%**
DeepSeek-R1-1.5B (1024T)	Qwen2.5-72B	15.2%	71.0%	83.2%
Qwen2.5-1.5B (Vanilla)	Qwen2.5-72B	15.2%	70.5%	**84.1%**
Qwen2.5-1.5B (Distilled)	Qwen2.5-72B	15.2%	**79.8%**	83.0%

As demonstrated in Table 2, deploying vanilla models yields suboptimal system accuracy. For instance, the vanilla Qwen2.5-1.5B achieves only 70.5% hybrid accuracy, heavily suffering from overconfident hallucinations. Even reasoning models like DeepSeek-R1 struggle in this constrained setting, achieving 71.0% accuracy while consuming more local energy.

However, by applying our routing-oriented distillation, the edge student learns to accurately assess its own uncertainty. Consequently, the system achieves a massive performance leap to 79.8% accuracy under a comparable energy budget.

5. Conclusion

In this paper, we proposed a Green-Edge-Cloud collaborative inference framework that bridges uncertainty estimation with system-level routing. By leveraging Routing-Oriented CoT Distillation, we successfully calibrated a 1.5B edge model to serve as an accurate, zero-shot confidence gatekeeper. Our rigorous hardware-level energy profiling demonstrates a superior Pareto frontier. By selecting the optimal knee point to offload 15.2% of queries, our system achieves 79.8% accuracy while reducing total energy consumption by 83%. Our results suggest that correctly ranking uncertainty, rather than perfectly calibrating probabilities, is sufficient to enable efficient and reliable edge-cloud collaboration.

Discussion and Future Work. While our empirical results on GSM8K demonstrate the clear dominance of confidence-calibrated routing, the absolute reliability of the intrinsic confidence signal can vary across different domains. We observe that for severe out-of-distribution (OOD) queries or non-reasoning factual recall tasks, the confidence–correctness alignment may become less stable. However, we expect the observed monotonic confidence–accuracy relationship to generalize to other reasoning benchmarks, as it arises from the intrinsic properties of autoregressive likelihood under calibrated training. Preliminary observations suggest that this trend is robust beyond GSM8K, although a systematic evaluation is left for future work.

Impact Statement

This paper aims to advance the field of machine learning by improving the energy efficiency of large language model inference. We believe that this research does not have any specific societal impacts that require special highlighting.

References

Bhattacharjee, P., Tian, F., Zhong, M., Zhang, G., Simeone, O., and Tandon, R. Conformal sparsification for bandwidth-efficient edge-cloud speculative decoding. *CoRR*, abs/2510.09942, 2025. doi: 10.48550/ARXIV.2510.09942.

Chen, L., Zaharia, M., and Zou, J. FrugalGPT: How to use large language models while reducing cost and improving performance. *Trans. Mach. Learn. Res.*, 2024, 2024.

Fan, H., Lin, Y.-C., and Prasanna, V. ELLIE: Energy-efficient LLM inference at the edge via prefill-decode splitting. In *2025 IEEE 36th International Conference on Application-specific Systems, Architectures and Processors (ASAP)*, pp. 139–146. IEEE, 2025. doi: 10.1109/ASAP65064.2025.00031.

Niu, C., Zhang, W., Zhao, Y., and Chen, Y. Energy efficient or exhaustive? benchmarking power consumption of llm inference engines. *ACM SIGENERGY Energy Informatics Review*, 5(2):56–62, 2025. doi: 10.1145/3757892.3757900.

Ong, I., Almahairi, A., Wu, V., Chiang, W.-L., Wu, T., Gonzalez, J. E., Kadous, M. W., and Stoica, I. RouteLLM: Learning to route LLMs with preference data. *CoRR*, abs/2406.18665, 2024. doi: 10.48550/ARXIV.2406.18665.

Panda, P., Magazine, R., Devaguptapu, C., Takemori, S., and Sharma, V. Adaptive llm routing under budget constraints. pp. 23934–23949, 2025.

Samsi, S., Zhao, D., McDonald, J., Li, B., Michaleas, A., Jones, M., Bergeron, W., Kepner, J., Tiwari, D., and Gadepally, V. From words to watts: Benchmarking the energy costs of large language model inference. In *IEEE High Performance Extreme Computing Conference, HPEC 2023, Boston, MA, USA, September 25-29, 2023*, pp. 1–9. IEEE, 2023. doi: 10.1109/HPEC58863.2023.10363447.

Wang, C., Li, H., Zhang, Y., Chen, L., Chen, J., Jian, P., Zhang, Q., and Hu, S. Icl-router: In-context learned model representations for llm routing. In *Fortieth AAAI Conference on Artificial Intelligence, Thirty-Eighth Conference on Innovative Applications of Artificial Intelligence, Sixteenth Symposium on Educational Advances in Artificial Intelligence, AAAI 2026, Singapore, January 20-27, 2026*, pp. 33413–33421. AAAI Press, 2026. doi: 10.1609/AAAI.V40I39.40628.

Xu, X., Feng, G., Liu, Y., Qin, S., Wang, J., and Wang, Y. Joint inference offloading and model caching for small and large language model collaboration. *IEEE Transactions on Mobile Computing*, 25(2):2691–2706, 2026. doi: 10.1109/TMC.2025.3608303.

Yuan, X., Li, H., Ota, K., and Dong, M. Generative inference of large language models in edge computing: An energy efficient approach. In *2024 International Wireless Communications and Mobile Computing (IWCMC)*, pp. 244–249. IEEE, 2024. doi: 10.1109/IWCMC61514.2024.10592339.

Zhang, C., Zhu, Z., Wei, Y., Tian, B., Liu, J., Wang, H., Xavier, W., and Liu, Y. Confidence-calibrated small-large language model collaboration for cost-efficient reasoning. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 4480–4501. Association for Computational Linguistics, 2026.